

One operating system for autonomous drone delivery in Dubai. Each order goes to the best-suited drone at the nearest pod. Drones fly geofenced routes and swap batteries at the Energy Hub on their own, and the Super Admin command center watches it all live.

9	1	36	≤30
SKYPODS	ENERGY HUB	DRONES	MIN SLA

The diagram illustrates the architecture of the UAE Air Traffic Control system, organized into four main sections:

- USERS & CHANNELS** (Top):
 - Customer app**: Order, pay, track live
 - Merchant portal & API**: Pharmacies, restaurants, labs
 - Super Admin command center**: Live 3D ops, fleet, airspace
 - Field-crew app**: Loading, recovery, maintenance
- Cloud Platform · UAE-HOSTED** (Middle):
 - Order & payments**: Intake, pricing, payments, alerts
 - Dispatch engine**: Nearest pod, payload, energy plan
 - Fleet & mission control**: Missions, re-planning, rebalancing
 - Energy manager**: Swap queue, packs, low-battery routing
 - Airspace & UTM**: Geofences, 90/120 m layers, separation
 - Telemetry & live video**: Position, battery, camera, flight logs
 - Analytics & reporting**: SLA, utilisation, regulator reports
 - Identity, roles & audit**: SSO, role-based access, audit trail

PostgreSQL + PostGIS · time-series telemetry store · MQTT / Kafka event bus · encrypted object storage
- FIELD NETWORK** (Bottom):
 - Drones × 36**: Autopilot (PX4 / OEM SDK), 4G/5G, Remote ID, ADS-B In, winch, 5 kg payload
 - SkyPods × 9**: Pod controller, 6 pads, weather sensor, parcel hand-off
 - Energy Hub × 1**: Robotic 60 s battery swap, charging racks, pack health
- EXTERNAL INTEGRATIONS** (Right):
 - UTM & Remote ID**: GCAA · Dubai Civil Aviation Authority
 - ADS-B traffic feed**: Manned aircraft & helicopters
 - Weather & dust alerts**: Wind, visibility, sandstorm holds
 - Maps & Makani**: Dubai addressing, geocoding
 - Payment gateway**: Cards, wallets, AED settlement
 - SMS · WhatsApp · push**: Order & delivery updates

Connectivity and Data Flow:

- Users & Channels connect to the Cloud Platform via **HTTPS / REST · WebSocket live updates**.
- The Cloud Platform connects to the Field Network via **MQTT telemetry & commands over 4G/5G · WebRTC video · Remote ID**.
- The Cloud Platform also interfaces with **EXTERNAL INTEGRATIONS**.

PHASE	TIMELINE (WEEKS)	SCOPE	EXIT MILESTONE
P0 Discovery & approvals path	W1-3 	Requirements & SLAs, hardware selection with OEM, site survey, GCAA / DCAA approval roadmap	Signed-off specification
P1 Platform core	W4-13 	Command center, dispatch engine, fleet & energy managers, airspace service, customer app, digital-twin simulator	End-to-end run on simulator
P2 Hardware integration & pilot	W12-23 	Autopilot SDK, pod & hub controllers, live telemetry & video, Remote ID. Pilot with 1 hub, 2 pods, 6 drones	First real deliveries
P3 City rollout	W24-33 	All 9 pods & 36 drones, UTM & ADS-B feeds, merchant API, payments, security audit, 24/7 ops training	Commercial go-live - ≈ month 8
P4 Scale & optimise	W34 → ongoing 	Swap stations at edge pods, demand forecasting, predictive maintenance, new cities	SLA-based support

- ✓ Nearest-pod dispatch with decision trace
- ✓ Automatic low-battery routing to the hub
- ✓ Geofence routing around DXB, DWC, Zabeel
- ✓ Fault & sandstorm auto-recovery
- ✓ Full-mission battery feasibility check
- ✓ Robotic swap queue & fleet rebalancing
- ✓ Live TFRs, separation, traffic advisories
- ✓ Customer app with live drone tracking

- Hardware from a selected OEM; we integrate the software side.
- The operator holds flight approvals; we supply the technical files.
- All data hosted in a UAE cloud region.
- Timelines are indicative until the quotation.

Intagrit · intagrit.com · Indicative plan for discussion. Final scope, timeline and commercials follow in the quotation.